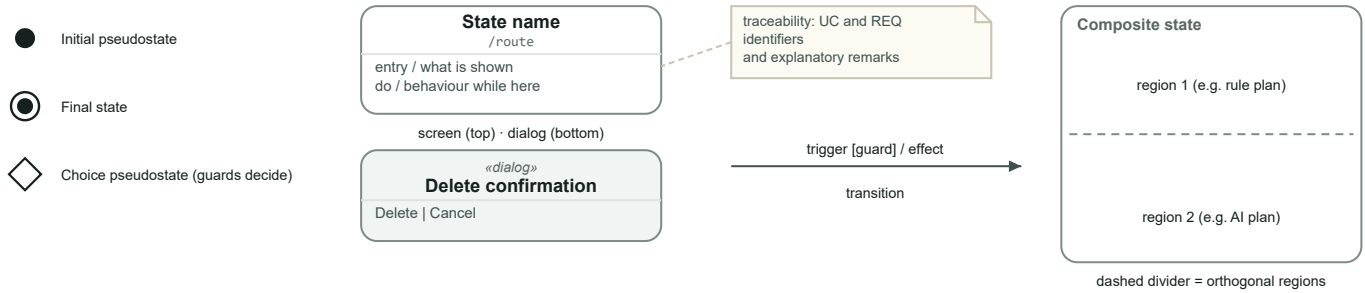
PathSG Dialog Map

Every screen, sub-state, dialog and transition in the PathSG interface, drawn as UML state machine diagrams -- one per feature area. Each state is something the user sees; each transition is labelled *trigger [guard] / effect*, and traceability to the use cases and requirements is carried in attached UML notes.

UC IDs refer to the Use Case Descriptions (v1.3); REQ IDs to the functional requirements of the SRS (v1.4); PERF / SEC / QUAL / BR IDs to the non-functional requirements and business rules. This version describes the application as shipped. Screen names and routes are those of the client router (`client/src/router/index.ts`).



MAP	WHAT IT COVERS	USE CASES
M Master	Every screen and dialog on one page, with the transitions that cross between maps 0 to 5	all
0 Shell	Header, roles, route guards, login landing, token renewal and session expiry, logout	UC-2.2, 2.3
1 Directory	Tiers, keyword / course / type / distance filters, sort, paging, school profile, location map, bookmark control	UC-1.1, 1.2, 2.5 - 2.7
2 Account	Create account, log in (with the login rate limit), student profile	UC-2.1, 2.2, 2.4
3a Generate	Rule-based and AI plans in parallel; ranking priority, distance limit, About, maps, AI limit	UC-3.1
3b Saved plan	Save with overwrite confirmation, view, edit, delete	UC-3.2 - 3.4
4 Reviews	Read, write (word cap), edit, delete; guest gate	UC-4.1 - 4.4
5 Admin	Multi-file school dataset update with locked controls; dataset export	UC-5.1, 5.2

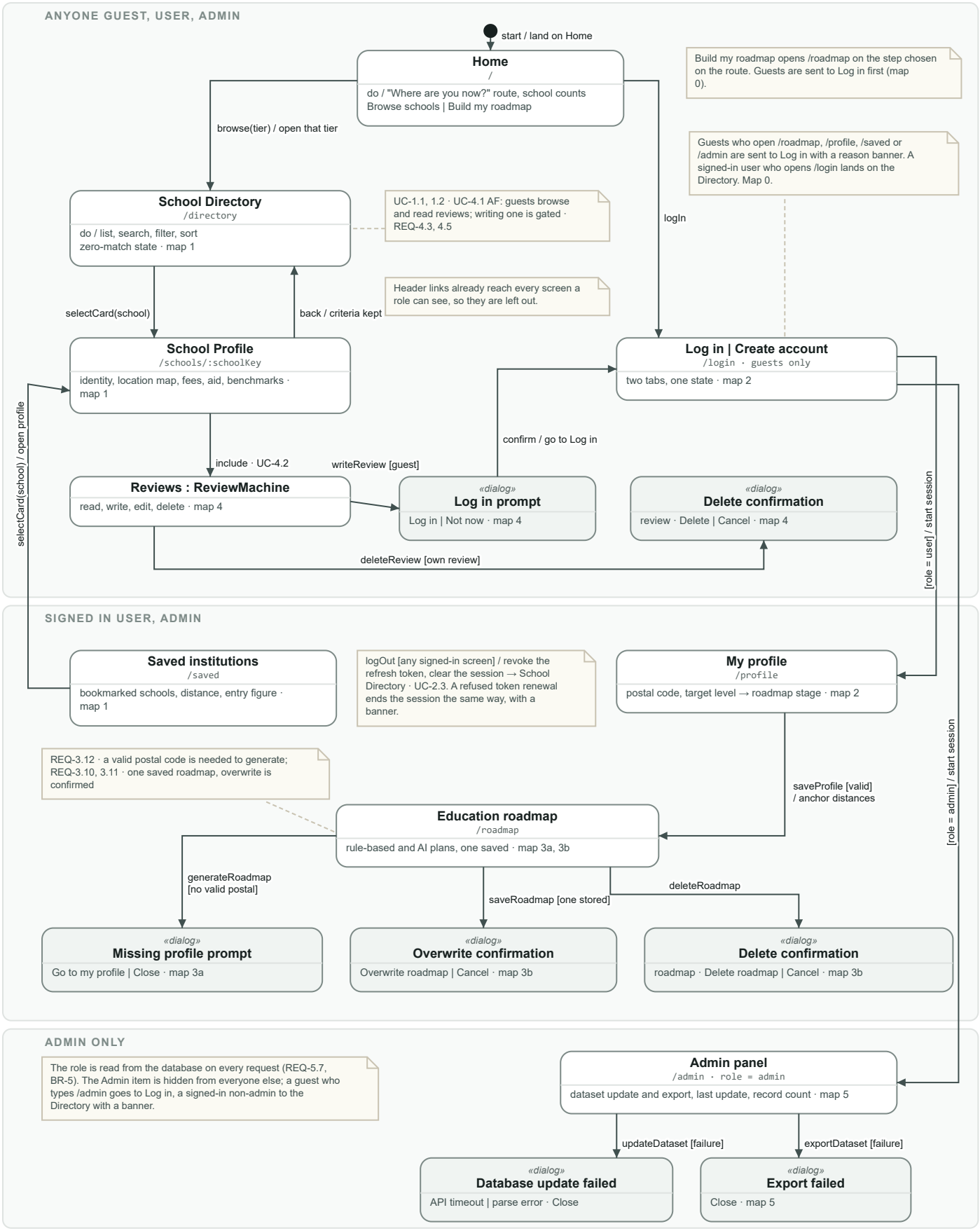
Changes from the Lab 3 version (v1.0, based on Use Case Descriptions v1.0)

- **Home screen (/)** is now where a visit starts, not the Directory. The "Where are you now?" route leads to the Directory or to the Roadmap.
- **Logout** ends at the Directory; a refused token renewal ends the session with a banner; the **login limit** (5 per 5 minutes) is shown as a rejected login.
- **Directory**: course filter (JC / Polytechnic / ITE and University), distance filter with its postal-code hints, keyword sanitising, load-failure and empty-tier states, the school **location map**.
- **Roadmap**: ranking priority (Rank by Grade or Distance), distance slider (re-ranks on release), About dialog, no Rule-Based Plan for Preschool and University, a map in every plan panel, the AI timeout of **15 s** (was 5 s) and rate limit, and an own Retry for the Rule-Based Plan.
- **Admin**: several CSV files per update, update limit of **10 s** (was 5 s), the "update already running" refusal, and the new **Export Database** flow (UC-5.2).
- **Errors**: dialogs are used for confirmations (overwrite, delete roadmap, delete review), the missing-postal and guest-review prompts, and failed admin updates and exports. Every other failure is an inline message or a banner, as drawn.

VERSION	DATE	CHANGE
1.0	Sep 2026	Initial dialog map (Lab 3)
2.0	7 Nov 2026	Aligned with SRS v1.4, Use Case Descriptions v1.3 and the shipped client

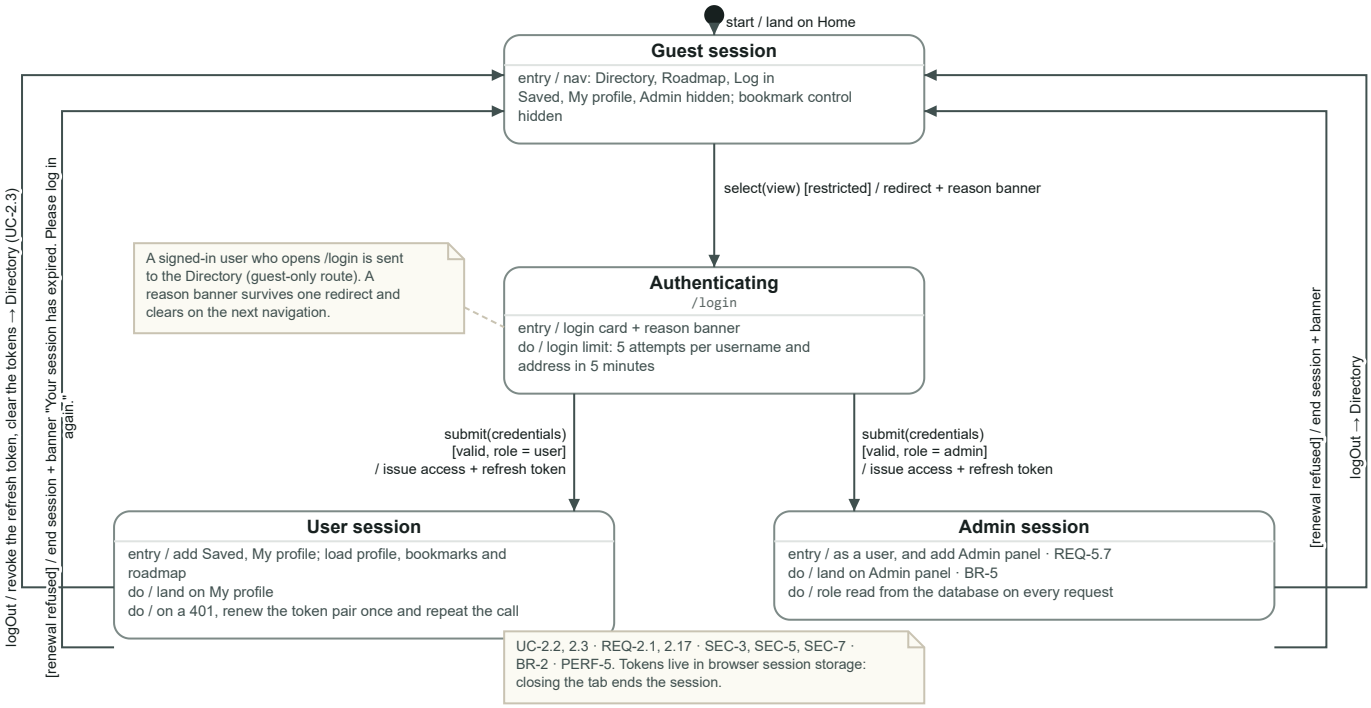
M Master dialog map

Every screen and dialog on one page, joined by the transitions that cross between maps 0 to 5. Bands group screens by who may open them. Each state names the map that expands it.



0 Shell and access control

One persistent header carries every map that follows. The role held by the session decides what it shows and where login lands. Directory and Roadmap are always present; Saved and My profile appear on login, and the Admin panel item only for the Admin role. Route guards, not the header, enforce access.

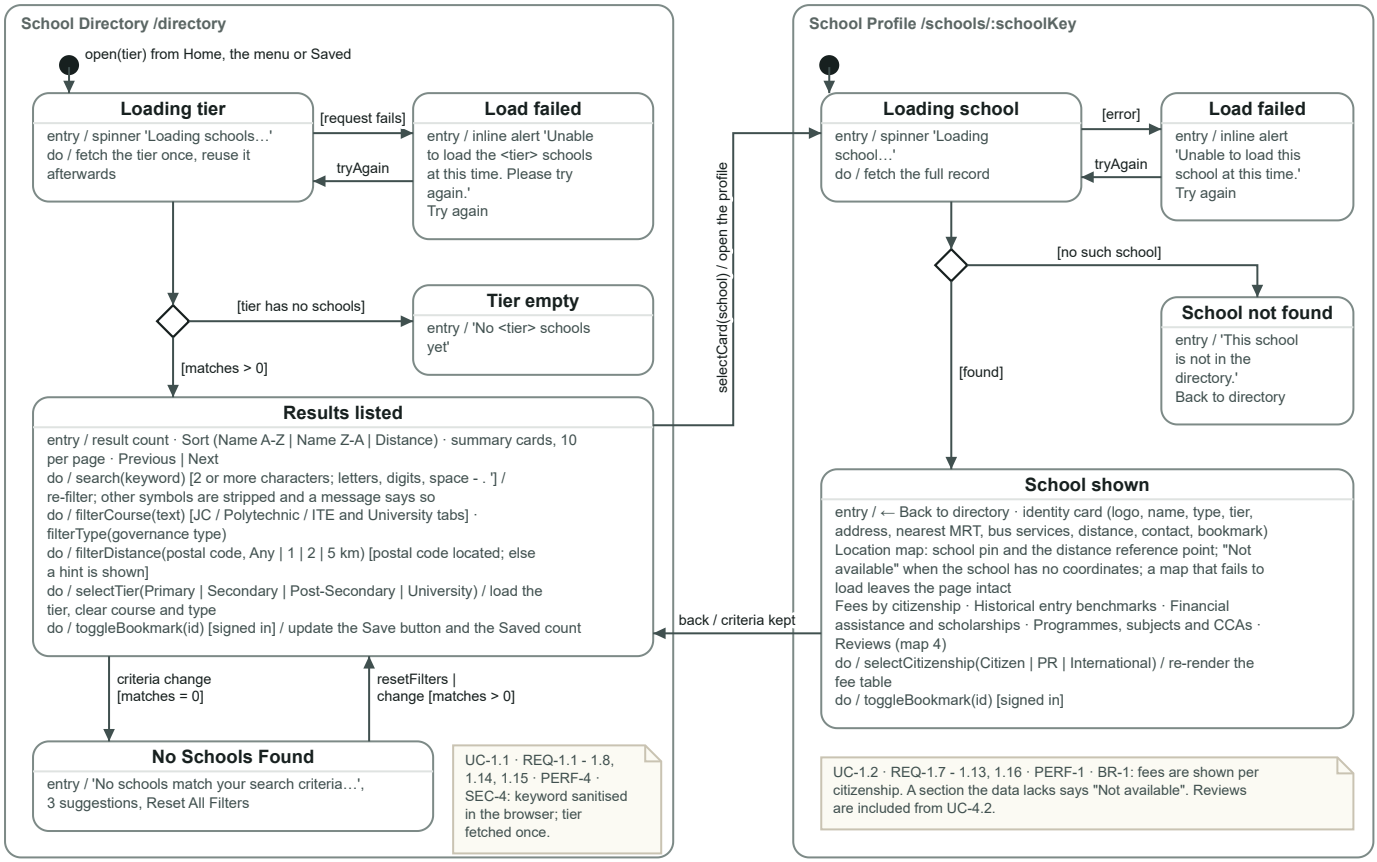


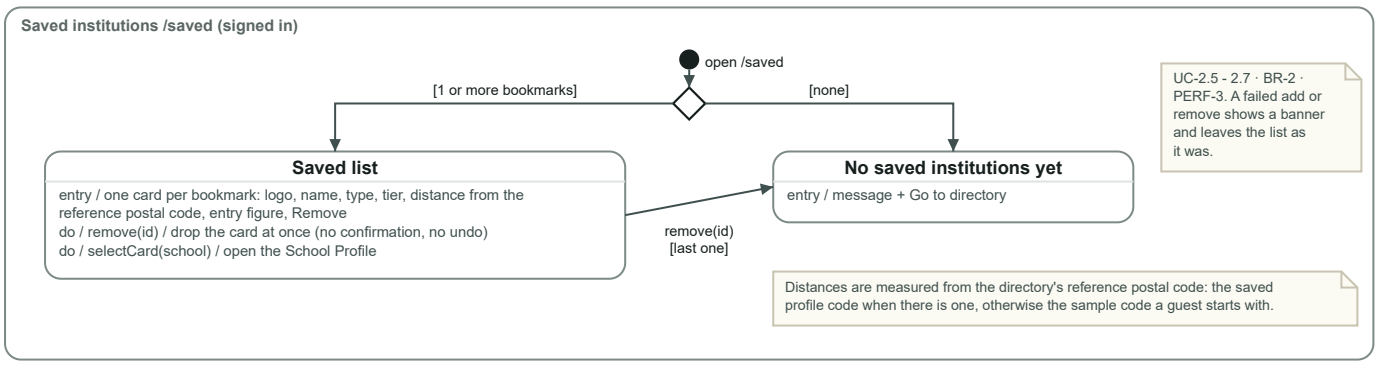
SCREEN OR ACTION	GUEST	USER	ADMIN	ENFORCED BY
Home, Directory, School profile	Full	Full	Full	REQ-4.3
Reviews: read	Full	Full	Full	REQ-4.3
Reviews: write, edit, delete	Gated by dialog	Own review only	Own review only	REQ-4.5, 4.10 · BR-2, BR-6
Bookmark control on cards and profile	Hidden	Full	Full	REQ-2.17 · BR-2
Roadmap, Saved, My profile	Redirect to Log in	Full	Full	REQ-2.17 · BR-2
Log in page	Full	Redirect to Directory	Redirect to Directory	guest-only route
Admin panel	Hidden; /admin goes to Log in	Hidden; /admin goes to Directory	Full	REQ-5.7 · BR-5

ROUTE	GUARD	MESSAGE SHOWN AFTER THE REDIRECT
`/roadmap`	requires login	"Log in to generate and save an education roadmap."
`/profile`	requires login	"Log in to edit your profile."
`/saved`	requires login	"Log in to view your saved institutions."
`/admin`	requires Admin role	Guest, on Log in: "Log in with an administrator account to open the admin panel." Signed-in user, on Directory: "The admin panel is restricted to administrator accounts."
`/login`	guests only	(a signed-in user goes to the Directory, no message)
any other path	none	(redirects to Home)

1 School Directory and profile

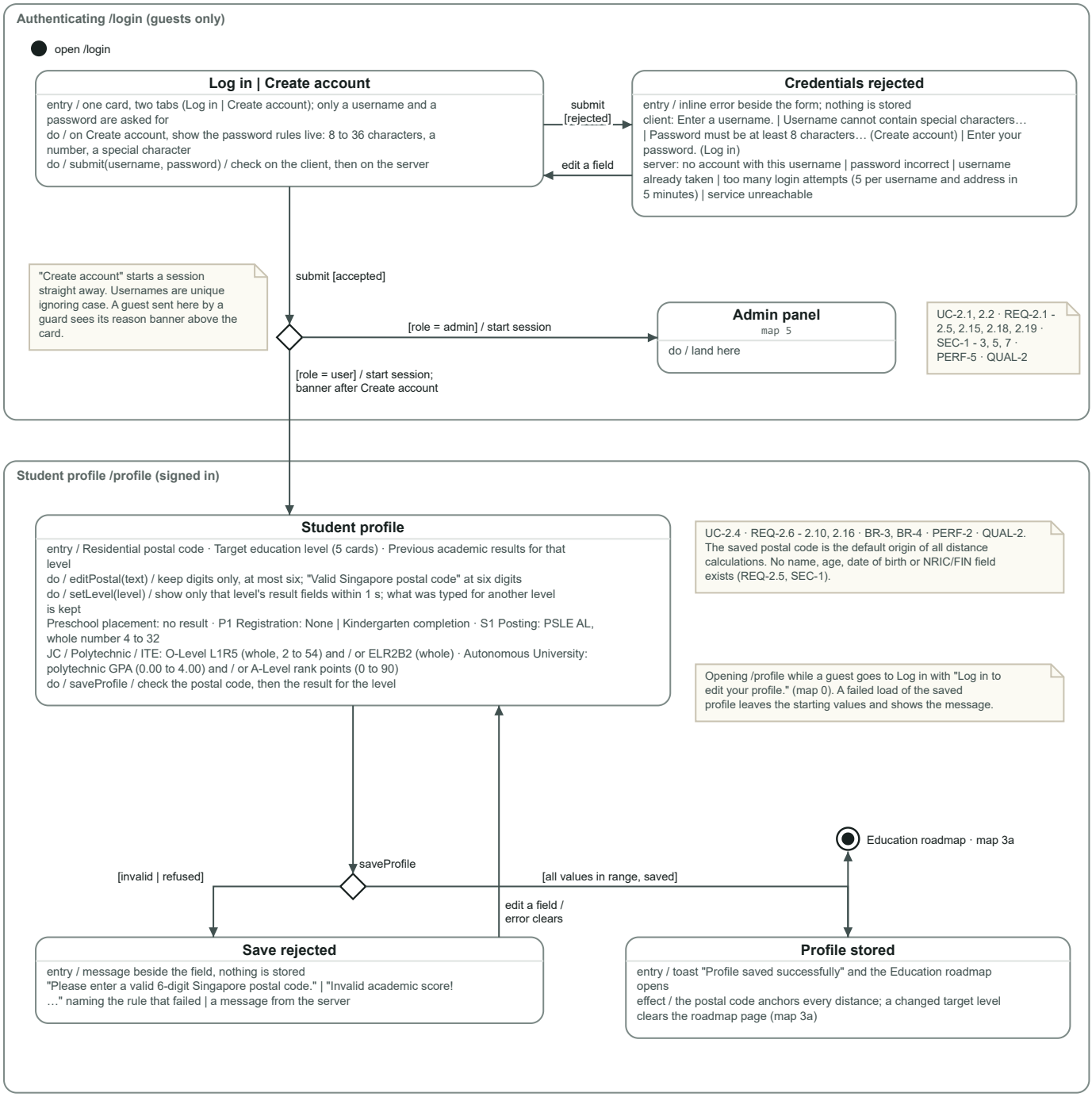
Search, filter and sort are internal transitions of one state, so the alternative flows (no match, malformed keyword, load failure) resolve through states and guards rather than separate pages. Selecting a card is the only transition that leaves the directory; back returns to the same tier, criteria, sort order and page, because the directory remembers them. The bookmark control exists only for signed-in users.





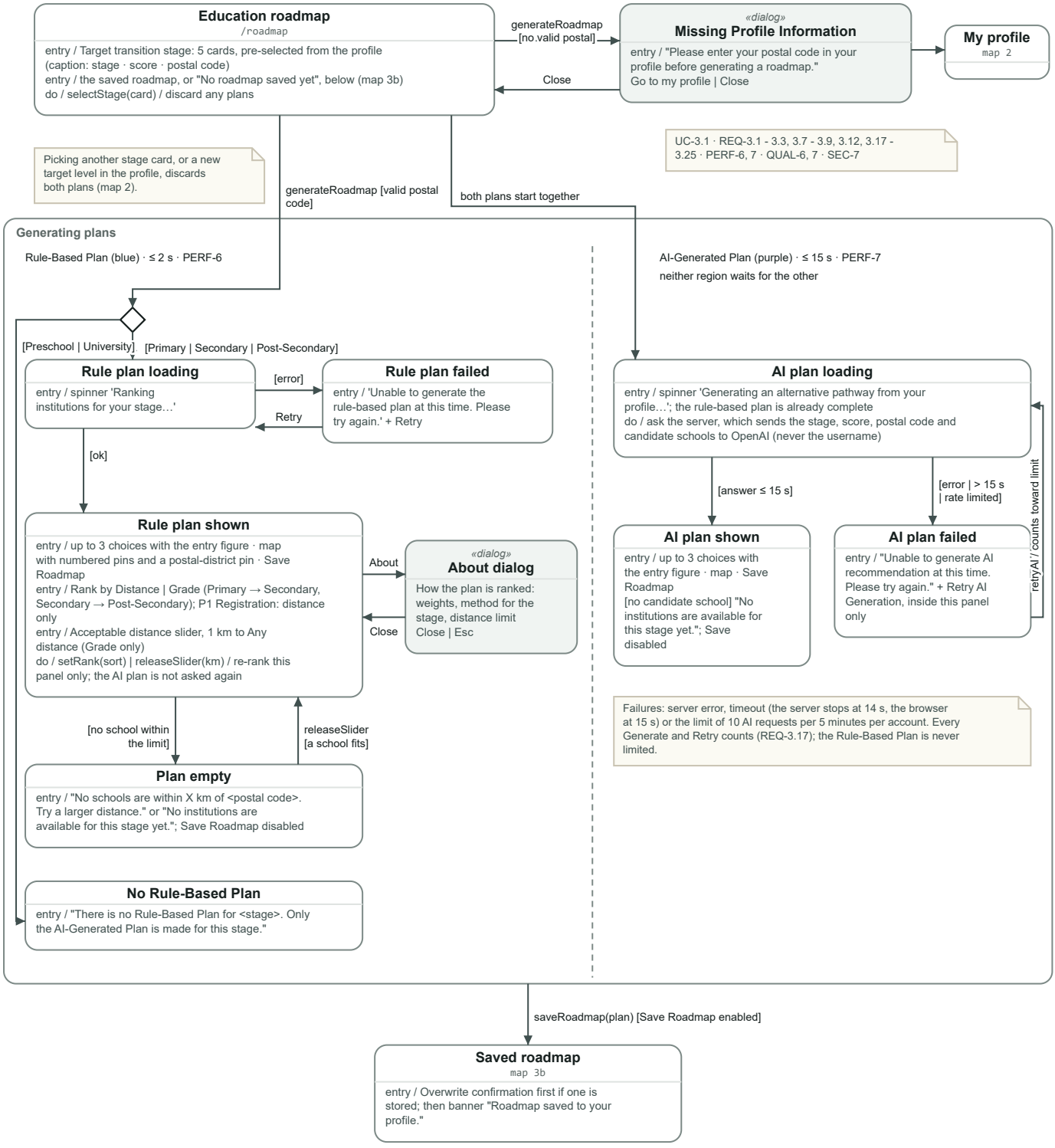
2 Account and student profile

Log in and Create account are two tabs of one state, so both paths reach the same session. The profile is the data spine of the system: its postal code anchors every distance and its target level pre-selects the roadmap stage, and neither can be stored out of range. Rejected input stays on the form; only a valid, accepted submission moves on.



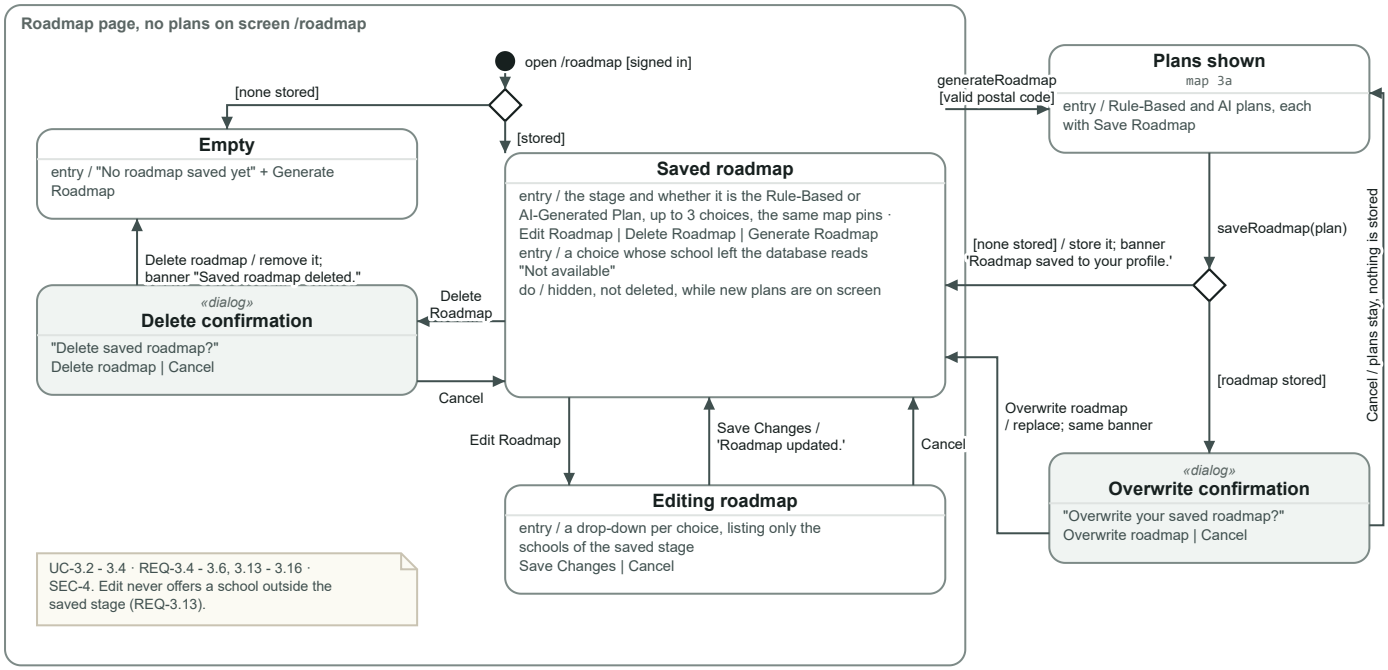
3a Roadmap generation

Generation is modelled as a composite state with two orthogonal regions, because that is the guarantee the requirements ask for: the rule-based region completes on its own clock, and any fault in the AI region stays inside that region with a retry of its own. Moving the ranking controls re-ranks only the rule-based region. Saving continues in map 3b.



3b Saving, editing and deleting a roadmap

One roadmap may be stored per account, which makes the overwrite confirmation part of the ordinary save path rather than an edge case. Cancelling anywhere leaves the stored record untouched. Plans on screen hide the saved roadmap without deleting it.

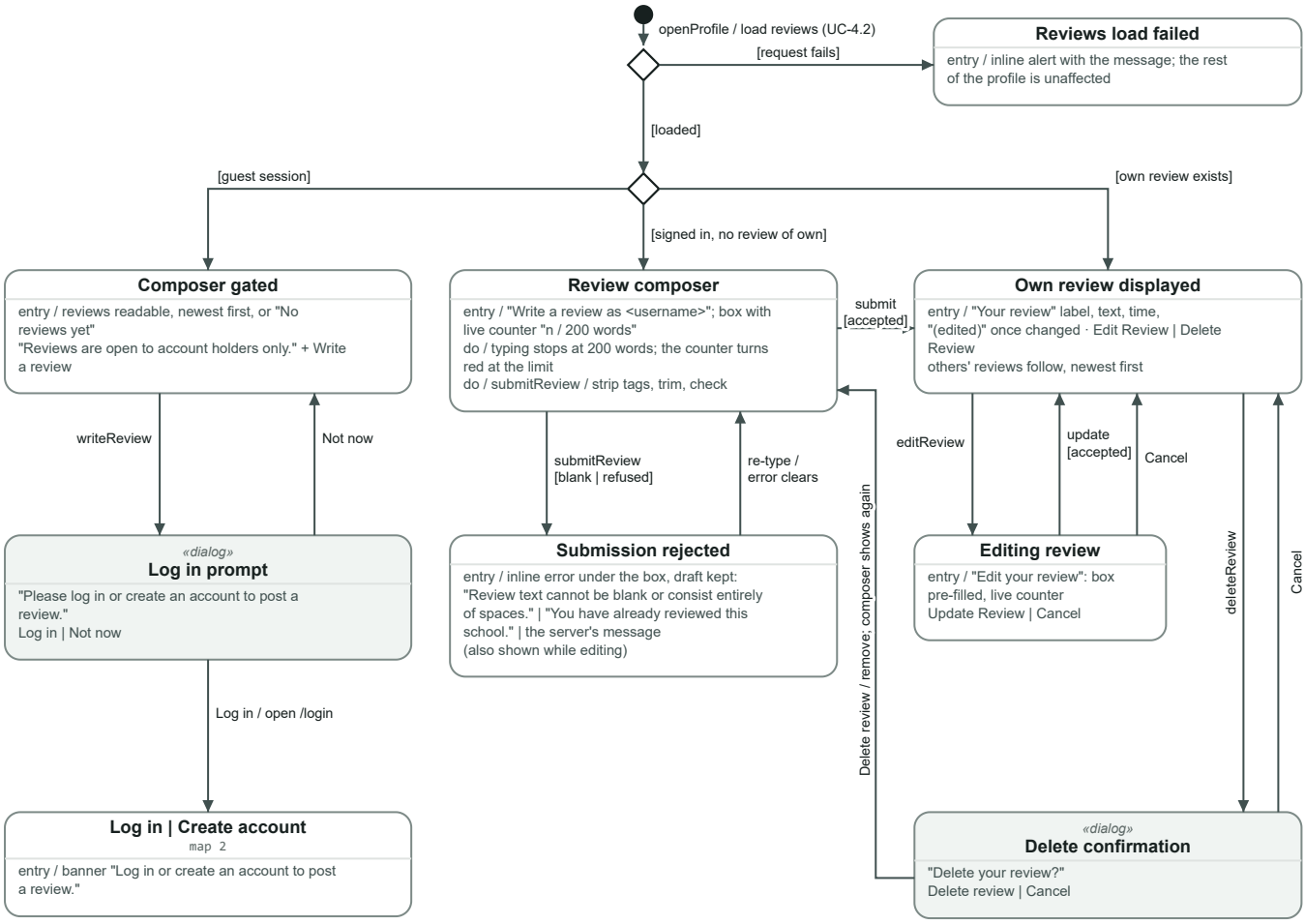


A failed save, edit or delete shows its message in the banner under the header and leaves the stored roadmap as it was. A banner clears when the user moves to another page.

One roadmap per account (REQ-3.10): saving always replaces. Overwrite is the only confirmation on the save path (REQ-3.11); delete is the only other one (REQ-3.15). Picking another stage card on /roadmap discards the plans and returns to this page.

4 School reviews

The submachine referenced by the school profile. Its first choice reads the session and the account's review history, so the same section opens gated, as a composer, or as the author's own card; editing and deleting are reachable only from that card.

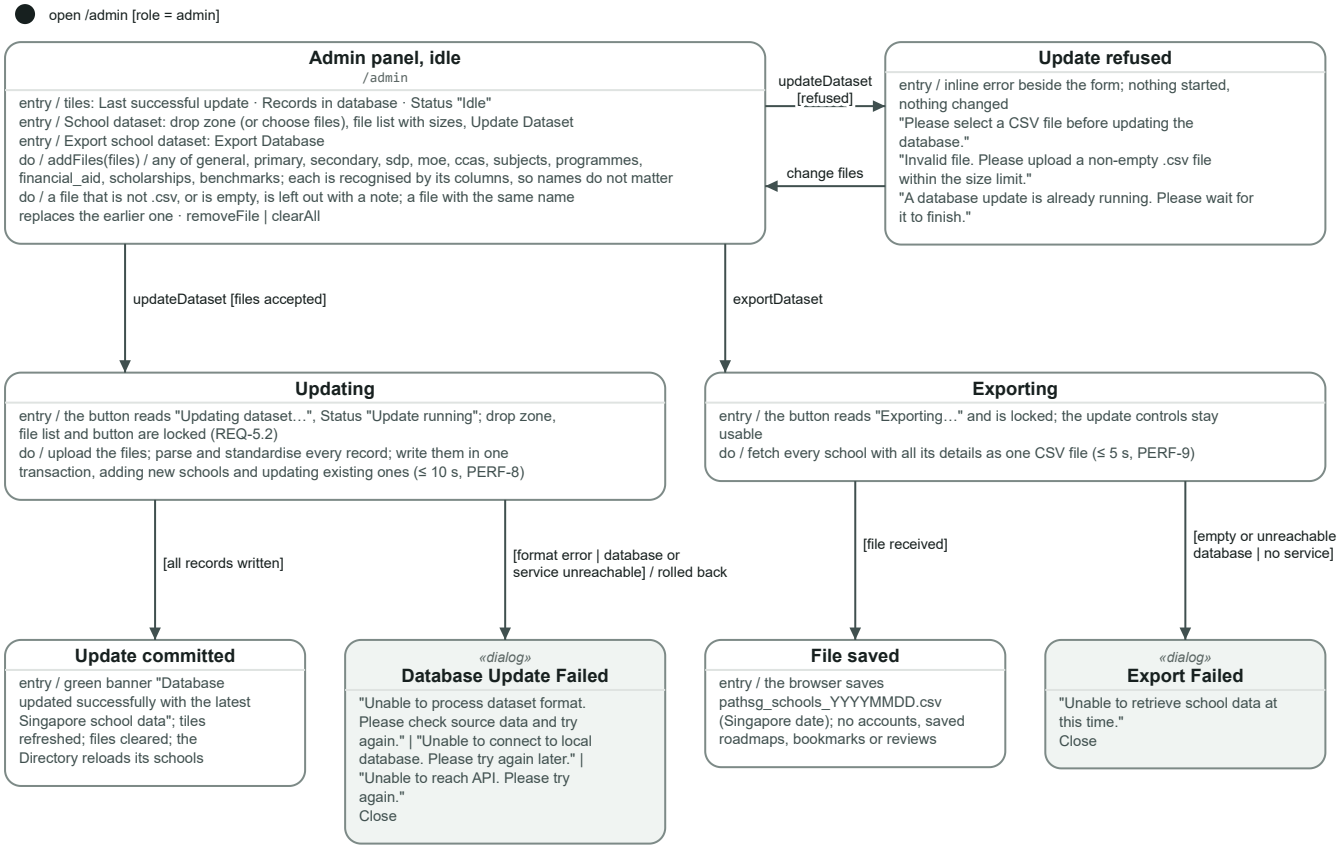


UC-4.1 - 4.4 · REQ-4.1 - 4.12 · BR-2, BR-6 · SEC-4 · QUAL-2. One review per account per school (BR-6): once posted, the box is replaced by the own review. Guests read everything; writing is gated by the dialog, never by a form that fails on submit.

The 200-word cap is enforced in the box and again by the server; HTML tags are stripped on both sides. A failed submit or edit stays inline in the box; a failed delete or load shows a banner or an inline alert. No review needs approval: it appears at once.

5 Admin: school dataset update and export

One panel, two independent triggers. Update locks its own controls for the run, and both failure branches leave the database as it was, so the panel always returns to a truthful idle state showing the last successful update. Export never touches the update lock.



Every outcome returns to "Admin panel, idle": the controls are enabled again; after a failure the database is unchanged.

UC-5.1, 5.2 · REQ-5.1 - 5.15 · BR-5 · SEC-6 · PERF-8, 9. Only an Admin account reaches this panel (map 0); the server re-checks the role on every request.

A kind of file that is left out keeps its last upload. Each file may be up to 20 MB (configurable). The update is all-or-nothing: any failure rolls everything back. Only one update runs at a time.